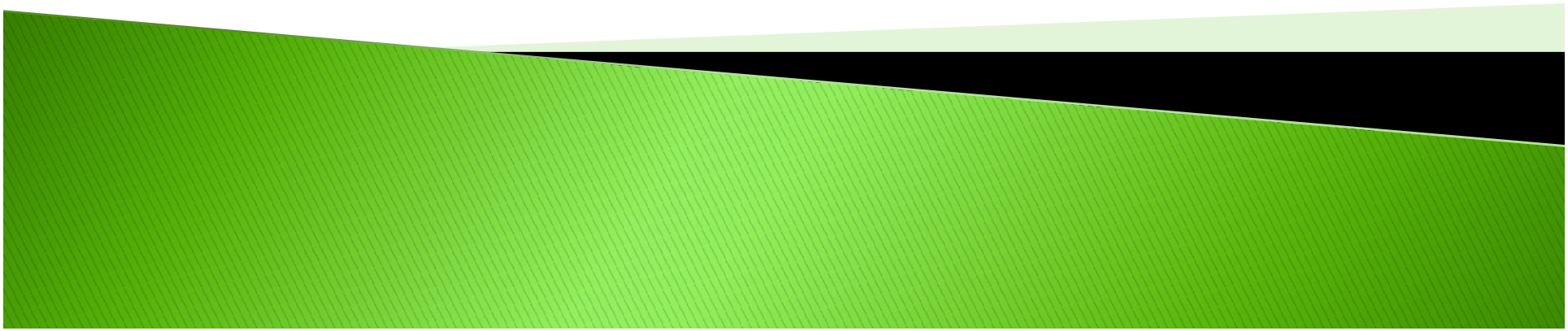
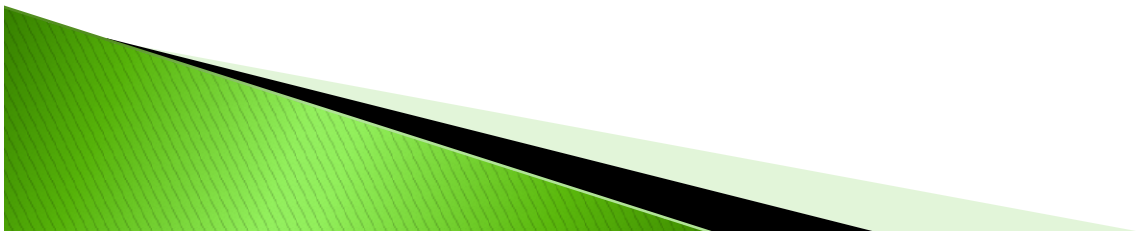
Sets and Logic



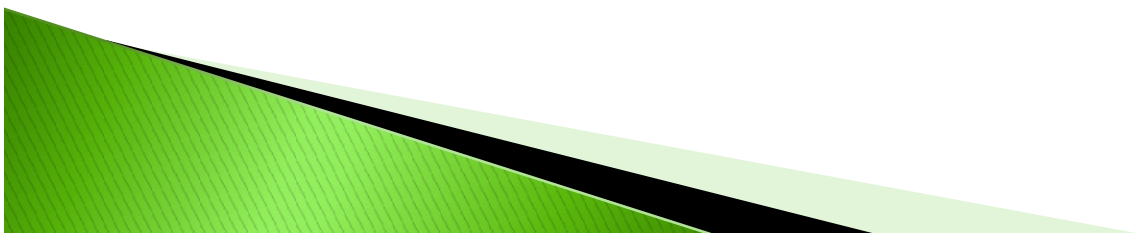
Introduction

- What is this course?
- Why do we study it?
- What is the relation between sets and logic?



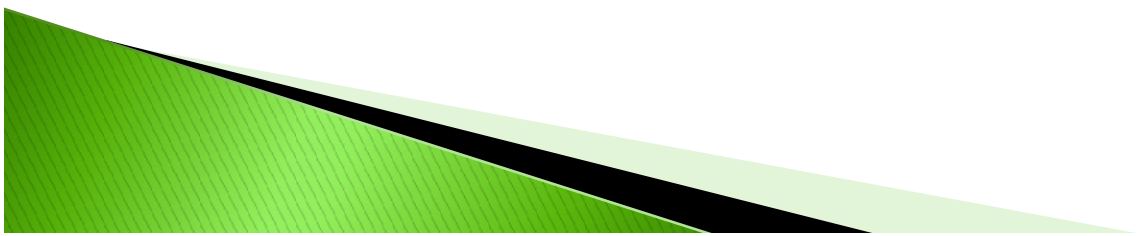
Introduction

In mathematics, a proof is a demonstration that, assuming certain axioms, some statement is necessarily true. That is, proof is a logical argument, not an empirical one. One must demonstrate that a proposition is true in all cases before it is considered a theorem of mathematics. An unproven proposition for which there is some sort of empirical evidence is a conjecture. Mathematical logic is the framework upon which rigorous proofs are built. It is the study of the principles and criteria of valid inference and demonstrations.



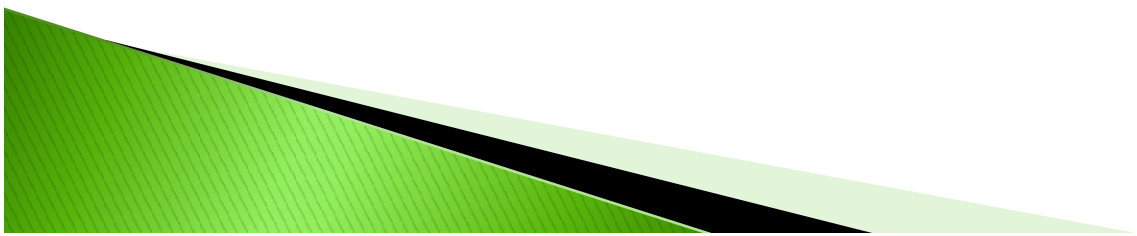
Introduction

In set theory, we study sets, which are informally collection of objects. Although any type of object can be collected into a set, set theory is applied most often to objects that are relevant to mathematics. A rigorous analysis of set theory belongs to the foundations of mathematics and mathematical logic. A Set Theorist explores what is provable—and what is not provable—when one is allowed to use all of the usual tools of mathematical constructions and proofs (tools such as: creating

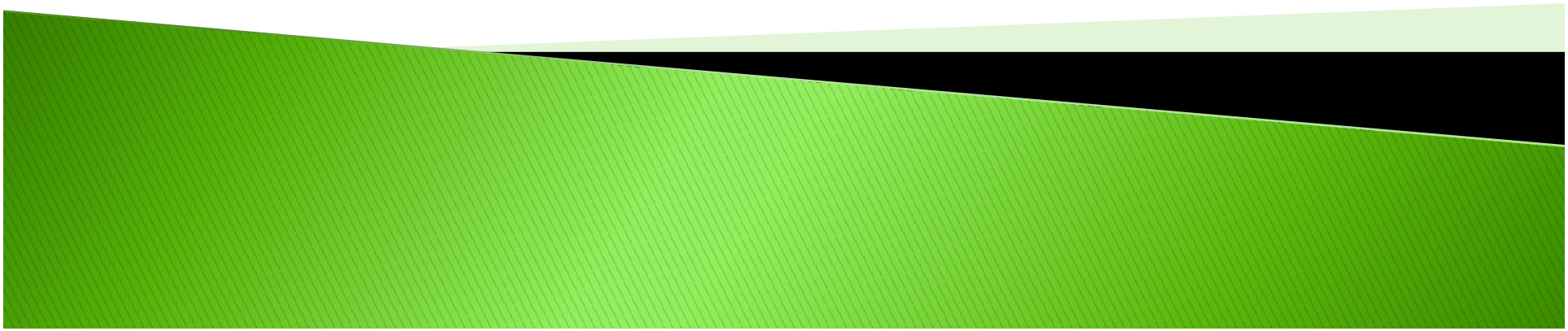


Introduction

function spaces, completions of various kinds of objects, taking quotients, forming products, induction and recursion, etc.).

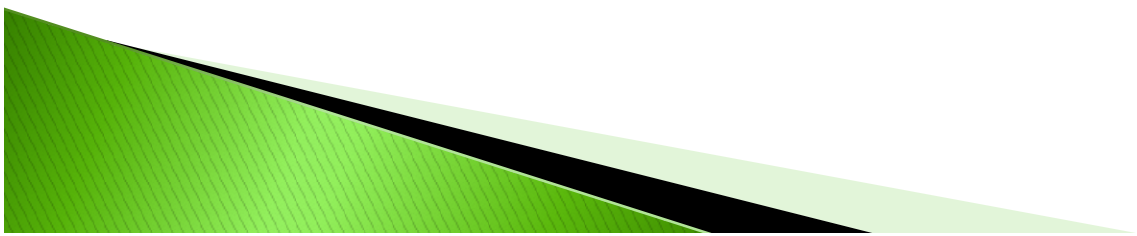


Basic Concepts of Sets



Sets and Elements

A fundamental concept in all branches of mathematics is that of a set. A set is any well-defined list, collection, or class of objects. The objects in sets, as we shall see from our examples, can be anything: numbers, people, letters, rivers, etc. These objects are called the elements or members of the set.



Some Examples of Sets

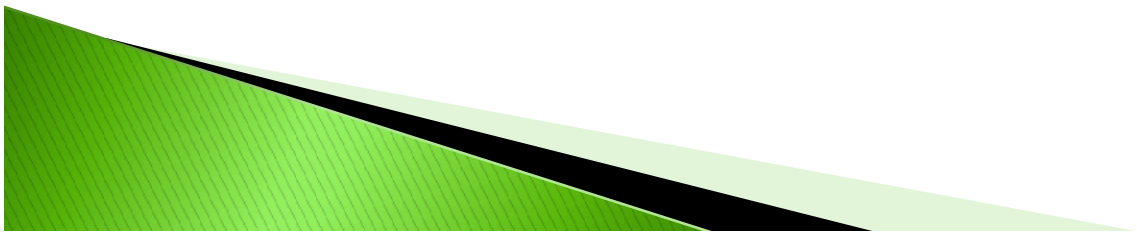
1) The numbers 1, 3, 7 and 10.

2) The solution of the equation $x^2 - 3x - 2 = 0$.

3) The vowels of the english alphabet:
a, e, i, o, u.

4) The people living on the earth.

5) The even integers 2, 4, 6, ...



Notation

Sets will usually be denoted by capital letters

A, B, X, Y,…

The elements in our sets will usually be represented by lower case letters

a, b, x, y,…

If we define a particular set by actually listing its members, for example, let A consists of the numbers 1,3, 7 and 10, then we write

$A = \{1, 3, 7, 10\}$

The statement "p belongs to A" is written as $p \in A$ .The statement that p is not an element of A, is written $p \notin A$

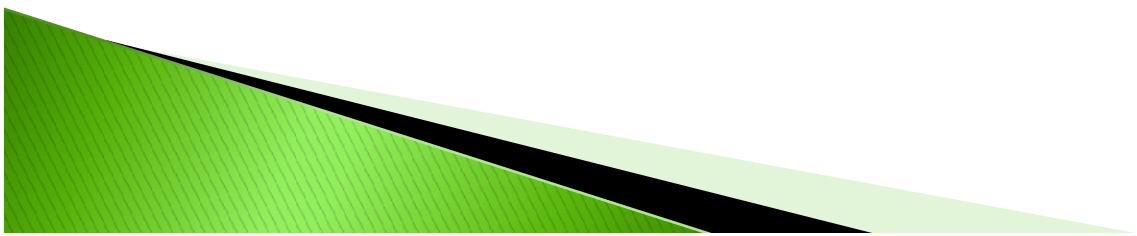
Notation

that is, the elements are separated by commas and enclosed in brackets { }. We call this the tabular form of a set.

But if we define a particular set by stating properties which its elements must satisfy, for example, let B be the set of all even numbers, then we use a letter, usually x, to represent an arbitrary element and we write

$$B = \{x \mid x \text{ is even}\}$$

We call this the set-builder form of a set.



Examples

1) Let $E = \{x \mid x^2 - 3x + 2 = 0\} = \{1, 2\}$

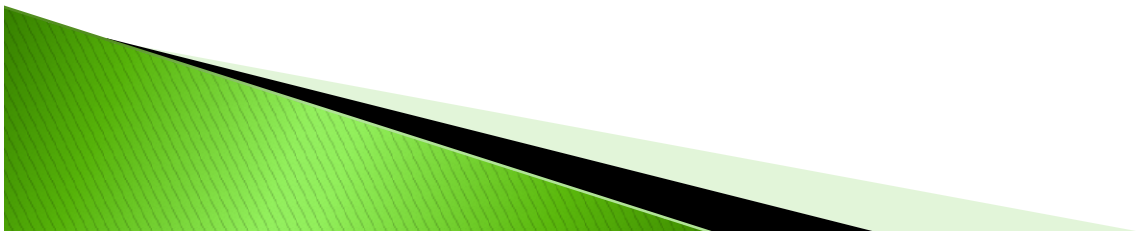
observe that $1 \in E$, $3 \notin E$.

2) $A = \{x \mid x \text{ is a vowel}\} = \{a, e, i, o, u\}$

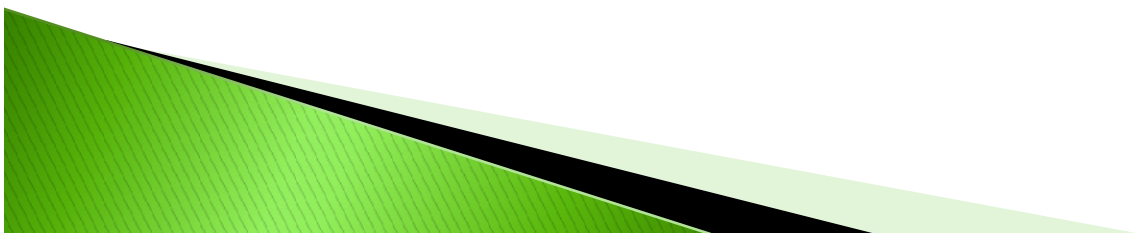
3) Let $F = \{x \mid x \in P, \underline{3 < x} \underline{< 12}\}$

where $P = \{1, 2, 3, 4, \ldots\}$

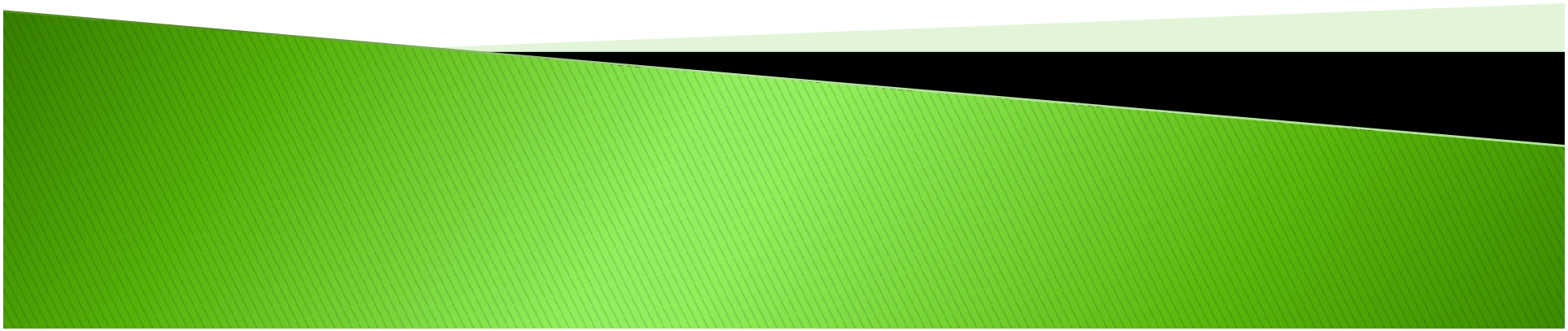
$F = \{4, 5, 6, 7, 8, 9, 10, 11\}$



Examples



Empty Set, Subsets and Universal Set

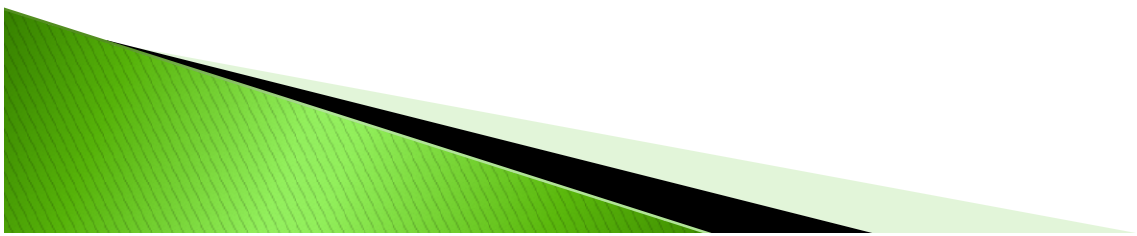


Universal Set

All sets under investigation in any application of set theory are assumed to be contained in some large fixed set called the universal set or universe. We will denote the universal set by U.

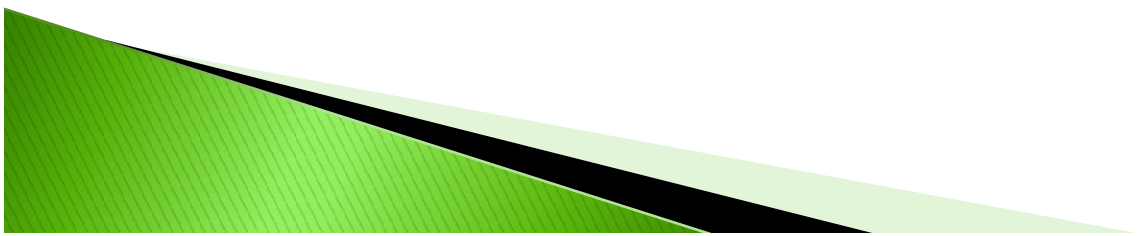
Examples are given below:

- In plane geometry, the universal set consists of all points in the plane.
- In human population studies the universal set consists of all people in the world.



Empty Set

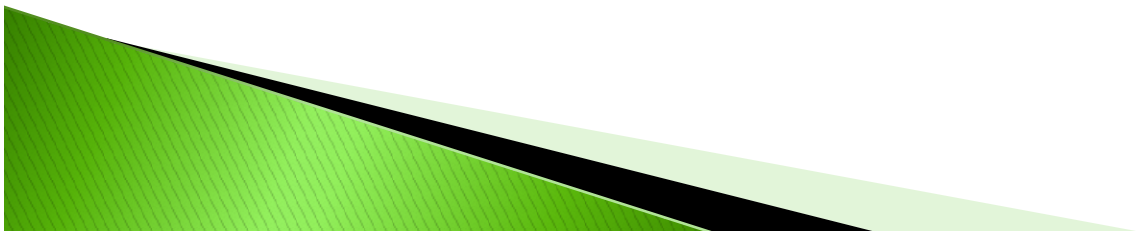
It is convenient to introduce the concept of the empty set, that is, a set which contains no elements. This set is sometimes called the null set. We denote it by symbol ø.



Some Examples

1. Let A be the set of people in the world who are older than 200 years.
2. Let $B = \{x \mid x^2 = 4, x \text{ is odd}\}$. Then B is the empty set.
3. The set $S = \{x : x \text{ is a positive integer}, x^2 = 3\}$ has no elements since no positive integer has the required property.

$S = \emptyset$

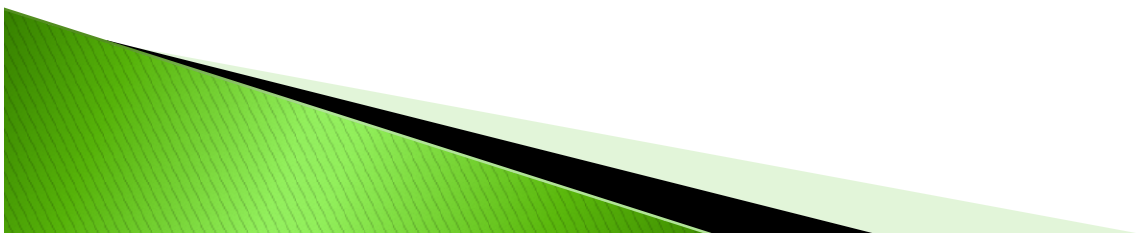


Subsets

Suppose every element in a set A is also an element of a set B ; then A is called a subset of B. We also say that A is *contained in* B or B *contains* A. This relationship is written

$$A \subseteq B \quad or \quad B \supseteq A$$

$A = \{1, 2\}$, $B = \{1, 2, 3\}$



Examples

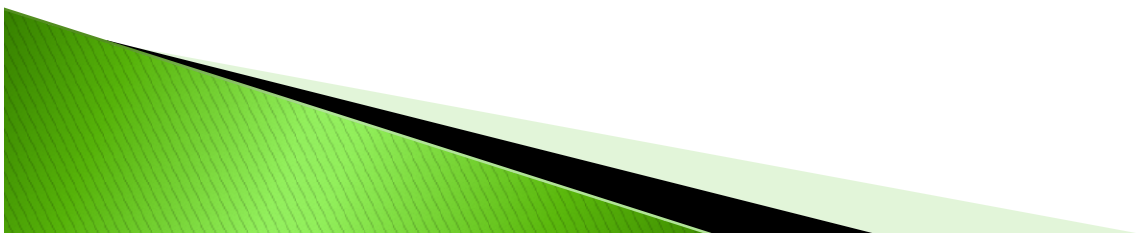
- Consider the sets

$$A=\{1, 3, 5, 8, 9\}, \quad B=\{1, 2, 3, 5, 7\}, \quad C=\{1, 5\}$$

Then $C \subseteq A$ and $C \subseteq B$. But $B \not\subset A$.

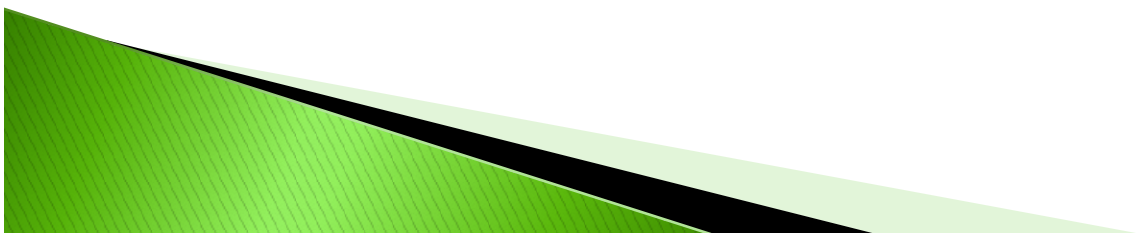
- The set $A = \{2,3,4,5\}$ is not a subset of $B = \{x : x \text{ is even}\}$ since $3 \in A$ but $3 \notin B$.

$= \{2, 4, 6, \ldots\}$



Properties of Sets

- Every set A is a subset of the universal set U since, by definition, all the elements of A belong to U. Also empty set ø is a subset of A.
- Every set A is a subset of itself since, trivially, the elements of A belong to A. $A \subseteq A$
- If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- If $A \subseteq B$ and $B \subseteq A$, then $A = B$.



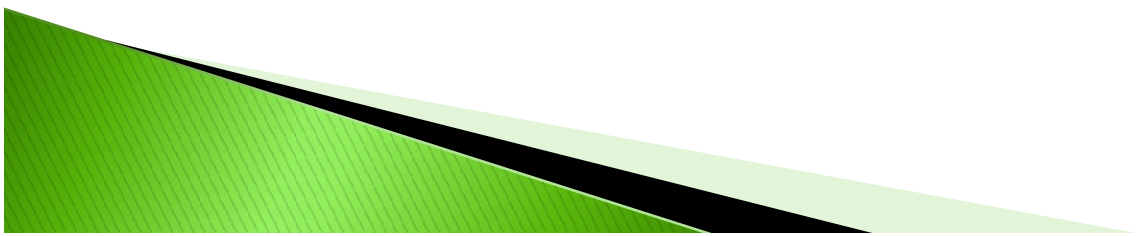
Proper Subset

If $A \subseteq B$, then it is still possible that $A = B$. When $A \subseteq B$ but $A \neq B$, we say that A is a proper subset of B.

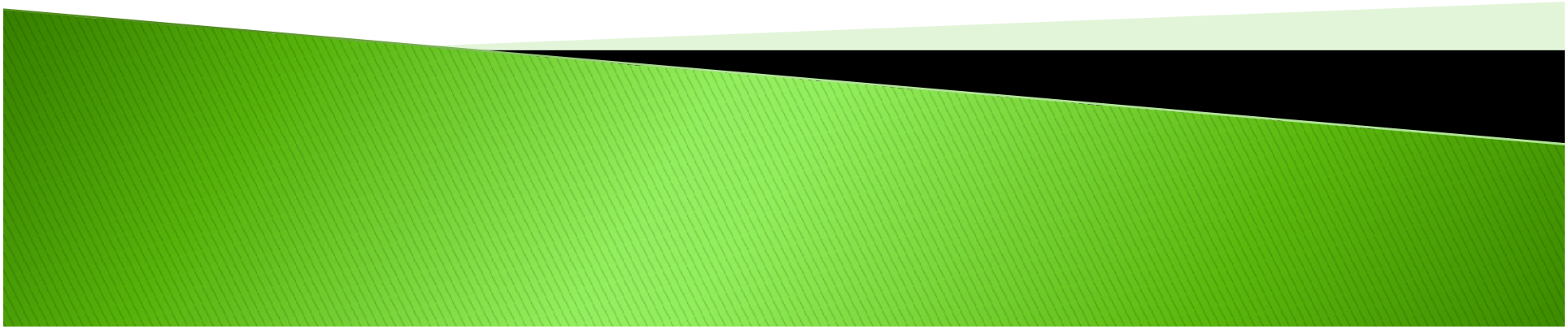
For example, suppose

$$A = \{1,2\}, \quad B = \{1,2,3\}, \quad C = \{1,3,2\}$$

Then A and B are both subsets of C; but A is a proper subset of C, whereas B is not a proper subset of C.

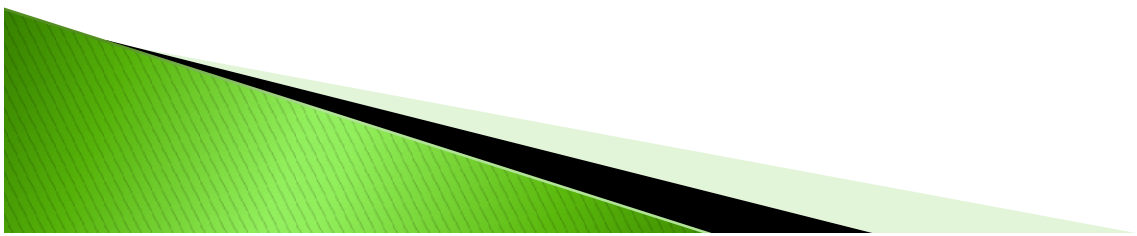


Set Operations



Set Operations

In this topic, we will study a number of set operations, including the basic operations of union, intersection, and difference of sets, where new sets will be assigned to pairs of sets A and B.



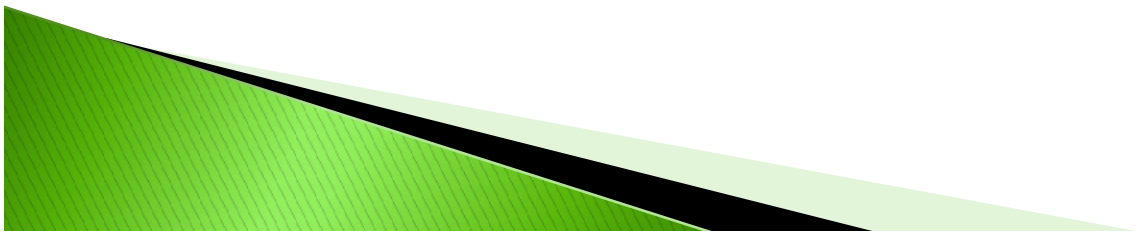
Union and Inersection

The union of two sets A and B, denoted by $A \cup B$, is the set of all elements which belong to A or B; that is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

The intersection of two sets A and B, denoted by $A \cap B$, is the set of all elements which belong to both A and B; that is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$



Example 1

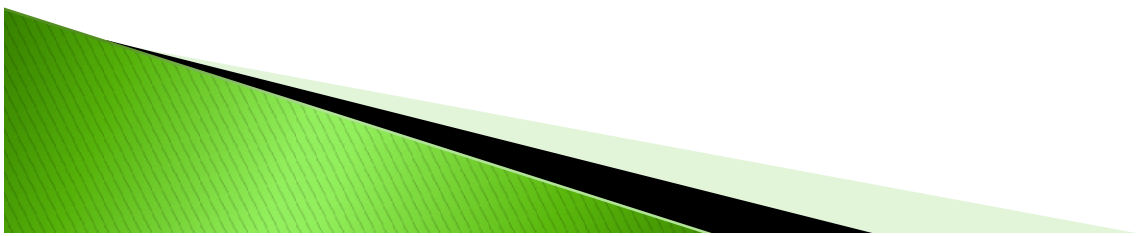
Let $A = \{1,2,3,4\}$, $B = \{3,4,5,6,7\}$, $C = \{2,3,8,9\}$.
Then

$A \cup B = \{1,2,3,4,5,6,7\}$, $A \cap B = \{3,4\}$

$A \cup C = \{1,2,3,4,8,9\}$, $A \cap C = \{2,3\}$

$B \cup C = \{3,4,5,6,7,2,8,9\}$, $B \cap C = \{3\}$

$= \{2,3,4,5,6,7,8,9\}$



Example 2

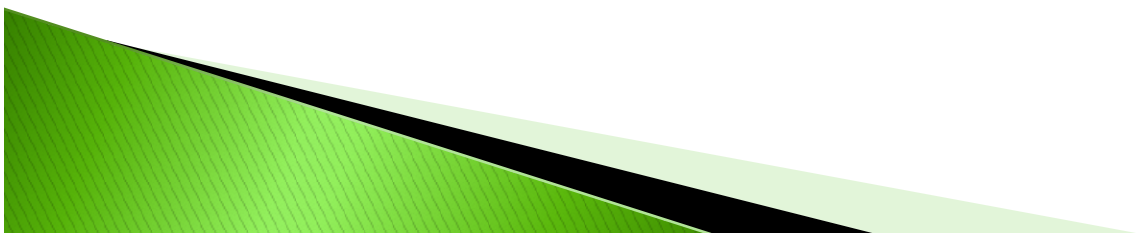
Let U denote the set of students at a university, and let M and F denote, respectively, the set of male and female students at the university. Then

$$M \cup F = U$$

since each student in U is either in M or in F . On the other hand,

$$M \cap F = \varnothing$$

since no students belong to both M and F.



Some Properties of Union and Intersection

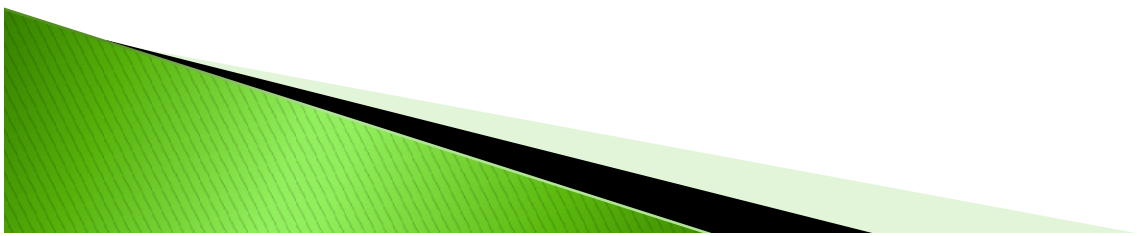
For any set A and B, we have

1) $A \cap B \subseteq A \text{ and } A \cap B \subseteq B$

2) $A \subseteq A \cup B \text{ and } B \subseteq A \cup B$

By combining the above two results, we can write

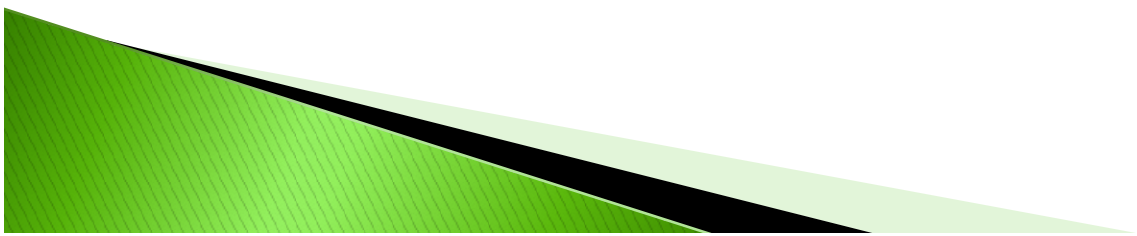
$$A \cap B \subseteq A \subseteq A \cup B \text{ and } A \cap B \subseteq B \subseteq A \cup B$$



Complement

Complement of a set A, denoted by A^c, is the set of elements which belong to U but which do not belong to A; that is,

$$A^c = \{x : x \in U, x \notin A\}$$



Example 3

Let $U = \{a, b, c, \ldots, y, z\}$ be the universal set, and let

$A = \{a,b,c,d,e\}$, $B = \{e, f, g\}$, $V = \{a,e,i,o,u\}$

Then

$A^c = \{f, g, h, \ldots, y, z\}$

$B^c = \{a, b, c, d, h, i, \ldots, y, z\}$

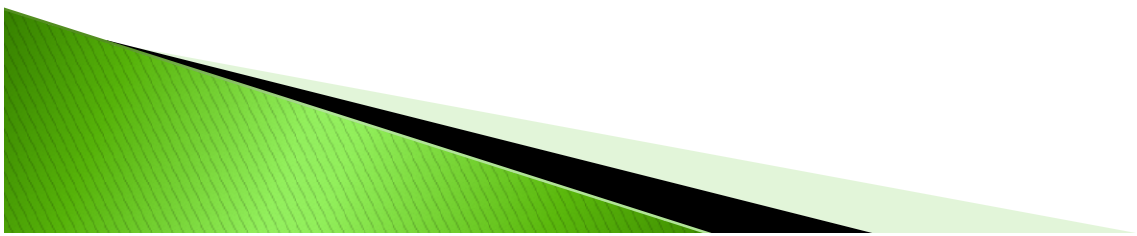
V^c = consonants

Example 4

Suppose the set R of real numbers is the universal set. $\mathbb{Q}$ denotes the set of rational numbers. Hence $\mathbb{Q}^c$ will denote the set of irrational numbers.

$Q = \{\frac{p}{q}, p \in Z, q \in Z, q \neq 0\}$ $\{\sqrt{2}, \sqrt{3}\}$ irrational

$Q^c = ?$

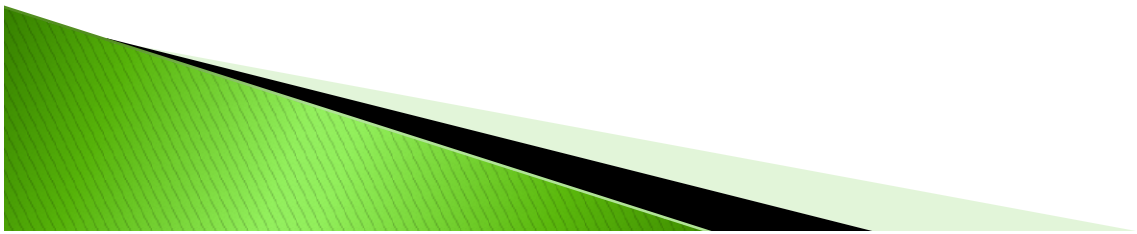


Difference of Sets

Let A and B be sets. The difference of A and B, denoted by A/B, is the set of elements which belong to A but which do not belong to B ; that is,

$$A / B = \{x : x \in A, x \notin B\}$$

Many texts denote A/B by A-B.



Example 5

Consider the sets

$A = \{1,2,3,4\}$, $B = \{3,4,5,6,7\}$, $C = \{6,7,8,9\}$

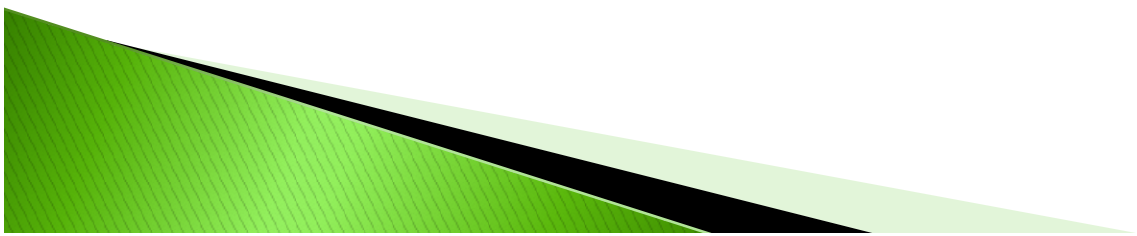
Then

$A/B = \{1,2\}$ ✓

$B/C = \{3,4,5\}$

$B/A = \{5,6,7\}$ ✓

$C/B = \{8,9\}$



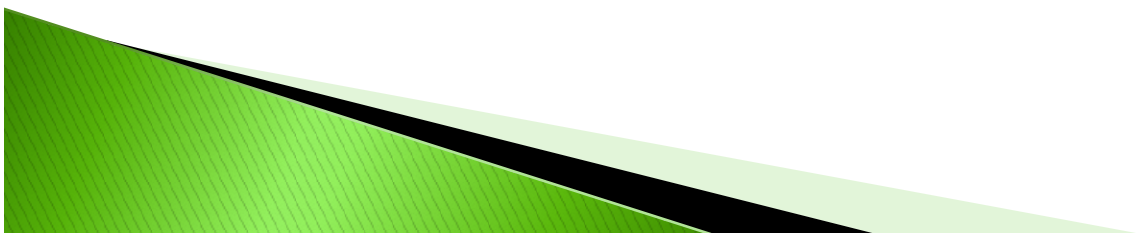
Symmetric Difference

The symmetric difference of the sets A and B, denoted by

$$A \oplus B = (A \cup B) / (A \cap B)$$

or

$$A \oplus B = (\mathrm{A}/\ \mathrm{B}) \cup (\mathrm{B}/\ \mathrm{A})$$



Example 6

Consider the sets

$A = \{1,2,3,4\}, \qquad B = \{3,4,5,6,7\}, \qquad C = \{6,7,8,9\}$

Then

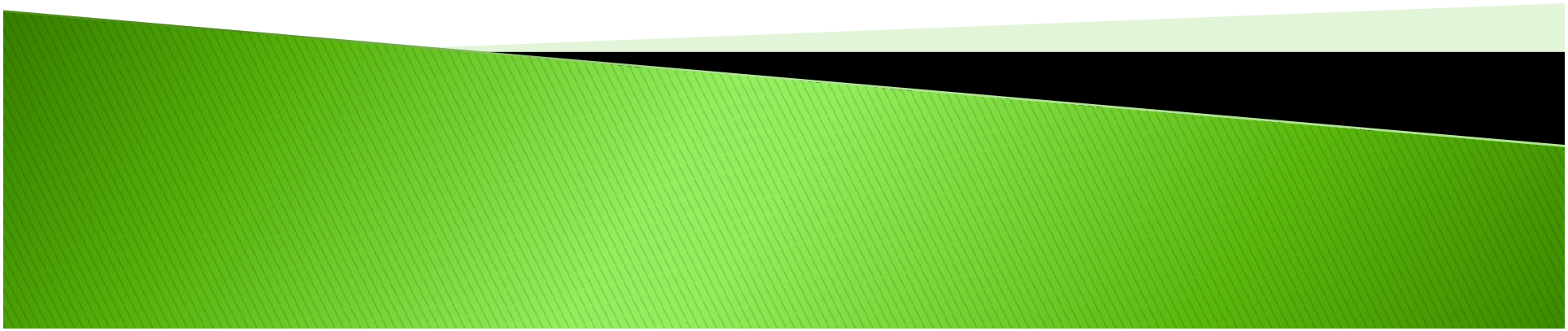
$A \oplus B = (A \cup B) / (A \cap B) = \{1,2,3,4,5,6,7\} / \{3,4\}$

$= \{1,2,5,6,7\}$

$B \oplus C = \{3,4,5,6,7,8,9\} / \{6,7\}$

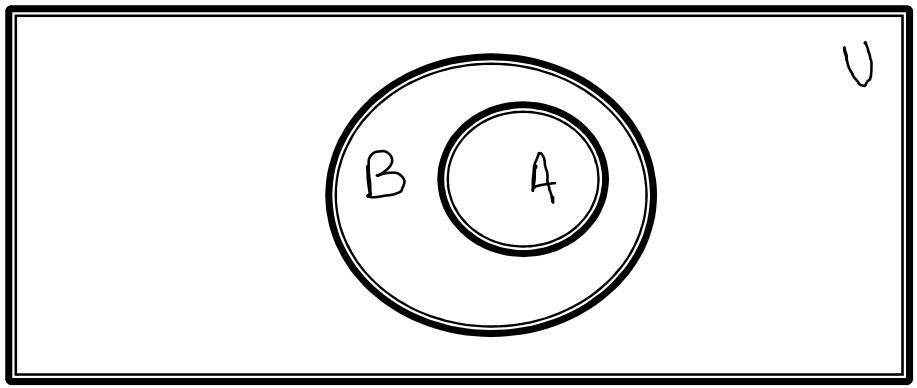
$= \{3,4,5,8,9\}$

Venn Diagrams



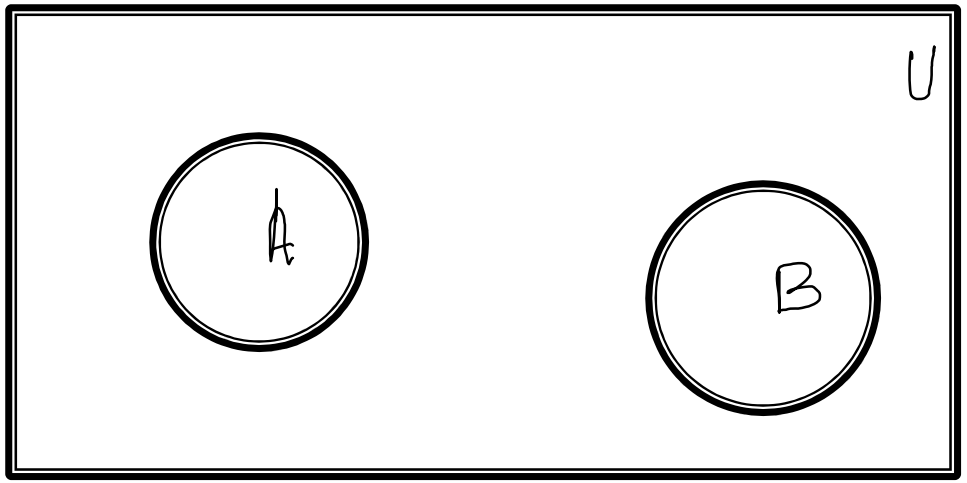
Venn Diagrams

A Venn diagram is a pictorial representation of sets where sets are represented by enclosed areas in the plane. The universal U represented by the points in a rectangle, and other sets are represented by disks lying within the rectangle. If $A \subseteq B$, then the disks representing A will entirely within the disk representing B, as shown in the following figure.



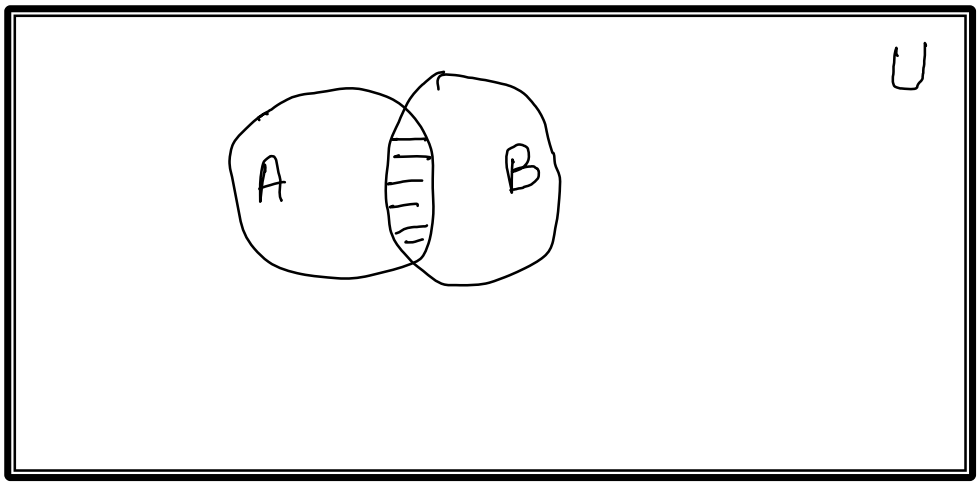
Venn Diagrams

If A and B are disjoint, then the disk representing A will be separated from the disk representing B, as in the following figure.



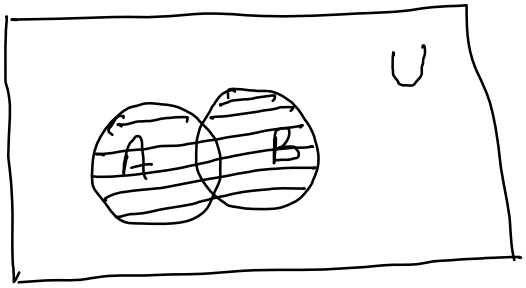
Venn Diagrams

On the other hand, if A and B are two arbitrary sets, it is possible that some elements are in A but not in B, some elements are in B but not in A, some are in both A and B, and some are in neither A nor B; hence , in general, we represent A and B as in the following way.

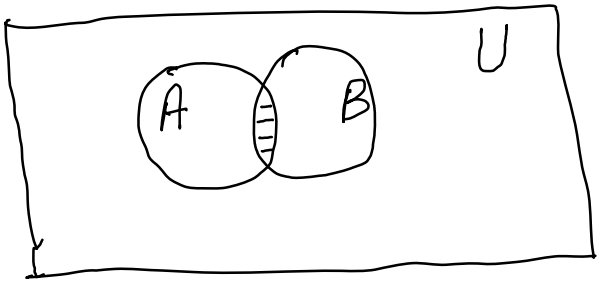


Some Examples

1) Union of A and B

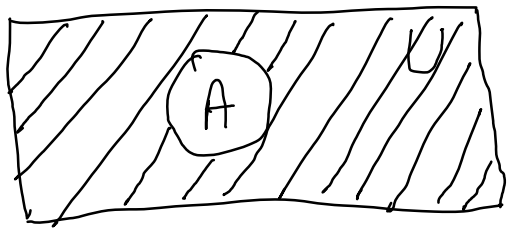


2) Intersection of A and B



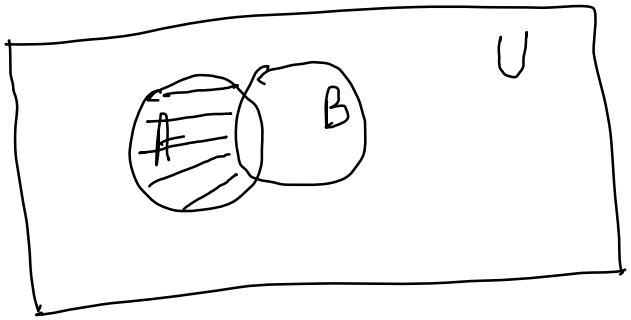
Examples

3) Complement of A



$A^c = U - A$

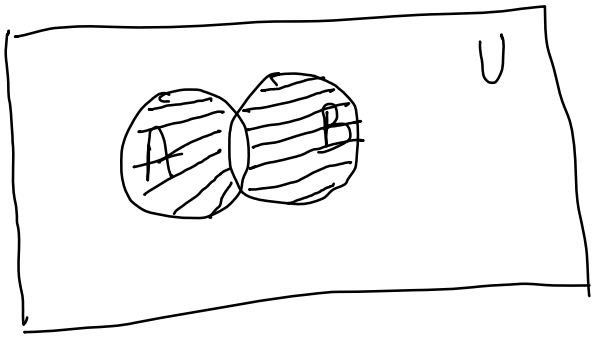
4) Difference of A and B



$A - B = \{x \mid x \in A, x \notin B\}$

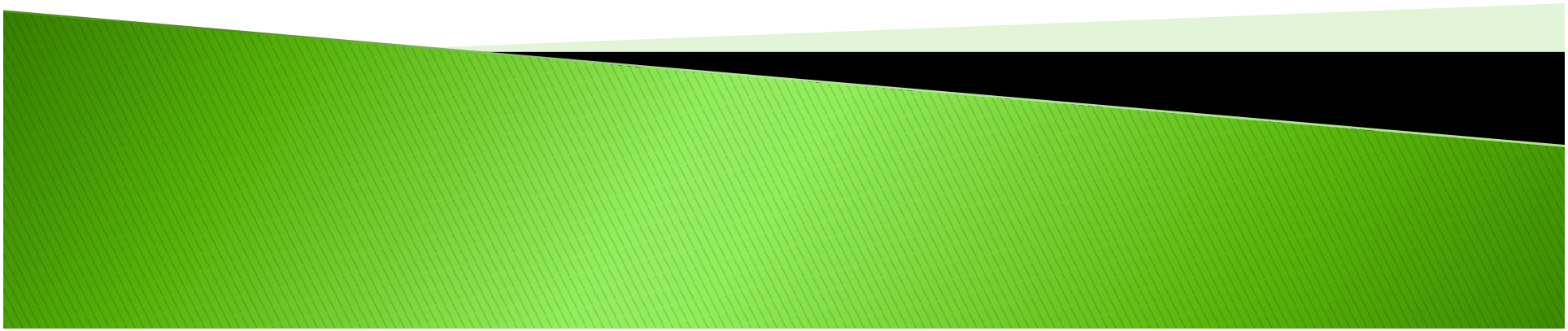
Examples

7) Symmetric difference of A and B



$$A \oplus B = (A \cup B) / (A \cap B)$$

Algebra of Sets, Duality



Laws of the Algebra of Sets

Sets under the operations of union, intersection and complement satisfy various laws(identities). Which are listed below one by one.

- Idempotent Laws

$$A \cup A = A \text{ and } A \cap A = A$$

$A \cup A = A$

L.H.S $= A \cup A$

$= \{x : x \in A \text{ or } x \in A\}$

$= \{x : x \in A\} = A$

$A \cap A = A$

L.H.S $= A \cap A$

$= \{x : x \in A \text{ and } x \in A\}$

$= \{x : x \in A\} = A$

Laws of the Algebra of Sets

- Associative Laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$

and

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$A \cup (B \cup C) = (A \cup B) \cup C$

L.H.S $= A \cup (B \cup C)$

$= \{x : x \in A \text{ or } x \in (B \cup C)\}$

$= \{x : x \in A \text{ or } x \in B \text{ or } x \in C\}$

$= \{x : x \in A \cup B \text{ or } x \in C\} = (A \cup B) \cup C$

Laws of the Algebra of Sets

- Commutative Laws

$$A \cup B = B \cup A$$

and

$$A \cap B = B \cap A$$

$A \cup B = \{x : x \in A \text{ or } x \in B\}$

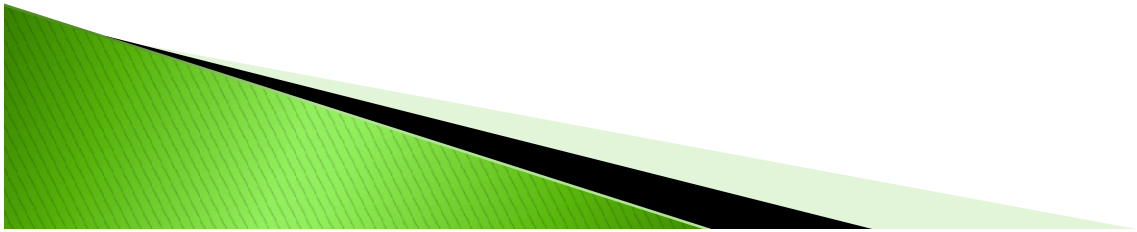
$= \{x : x \in B \text{ or } x \in A\}$

$= B \cup A$

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

$= \{x : x \in B \text{ and } x \in A\}$

$= B \cap A$



Laws of the Algebra of Sets

- Distributive Laws

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

and

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$L.H.S = A \cap (B \cup C)$

$= \{x : x \in A \text{ and } x \in B \cup C\}$

$= \{x : x \in A \text{ and } x \in B \text{ or } x \in A \text{ and } x \in C\}$

$= \{x : x \in (A \cap B) \text{ or } x \in (A \cap C)\}$

$= (A \cap B) \cup (A \cap C)$

Laws of the Algebra of Sets

- Identity Laws

$$A \cup \phi = A, \quad A \cap U = A$$

and

$$A \cup U = U, \quad A \cap \phi = \phi$$

$L.H.S = A \cup \phi = \{x : x \in A\} = A = R.H.S$

$R.H.S = A = \{x : x \in A\} = \{x : x \in A \text{ and } x \in U\}$

$= A \cap U$

Laws of the Algebra of Sets

- Involution Law

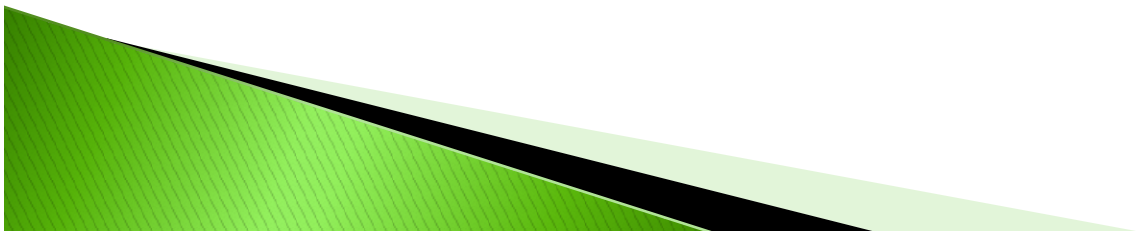
$$(A^c)^c = A$$

Note that $A^c = \{x : x \in U, x \notin A\}$

So, $(A^c)^c = \{x : x \in U, x \notin A^c\}$

$= \{x : x \in U, x \in A\}$

$= A$



Laws of the Algebra of Sets

- Complement Laws

$$A \cup A^c = U, \quad A \cap A^c = \phi$$

and

$$U^c = \phi, \quad \phi^c = U$$

1) $L.H.S = A \cup A^c$

$= \{x : x \in A \text{ or } x \in A^c\}$

$= \{x : x \in A \text{ or } x \in U, x \notin A\}$

$= \{x : x \in U\} = U = R.H.S$

2) $L.H.S = A \cap A^c$

$= \{x : x \in A \text{ and } x \in A^c\}$

$= \{x : x \in A \text{ and } x \in U, x \notin A\}$

$= \phi = R.H.S$

Laws of the Algebra of Sets

- DeMorgan's Laws

1) $(A \cup B)^c = A^c \cap B^c$

and

2) $(A \cap B)^c = A^c \cup B^c$

1) L.H.S $= (A \cup B)^c$

$= \{x : x \notin (A \text{ or } B)\}$

$= \{x : x \notin A \text{ and } x \notin B\}$

$= \{x : x \in A^c \text{ and } x \in B^c\}$

$= A^c \cap B^c = R.H.S$

Duality

The identities that we have discussed above are arranged in pairs. We now consider the principle behind this arrangement. Let E be an equation of set algebra. The dual E^* of E is the equation obtained by replacing each occurrence of $\cup, \cap, U, \varnothing$ in E by $\cap, \cup, \varnothing, U$, respectively.

For example, the dual of

$$(U \cap A) \cup (B \cap A) = A \text{ is } (\varnothing \cup A) \cap (B \cup A) = A.$$

dual of

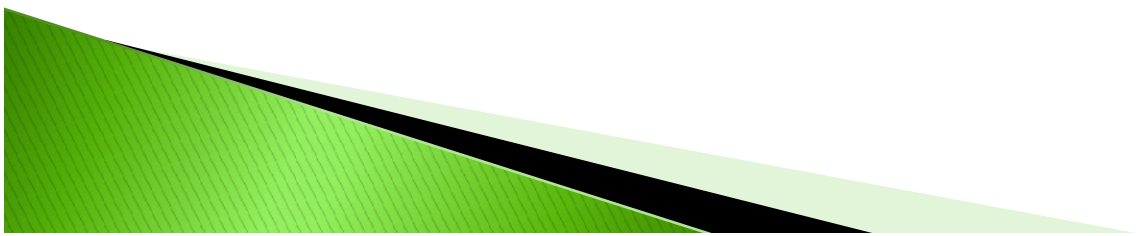
Example

Question. Write the dual $(A \cap U)^c \cap A = \varnothing$.

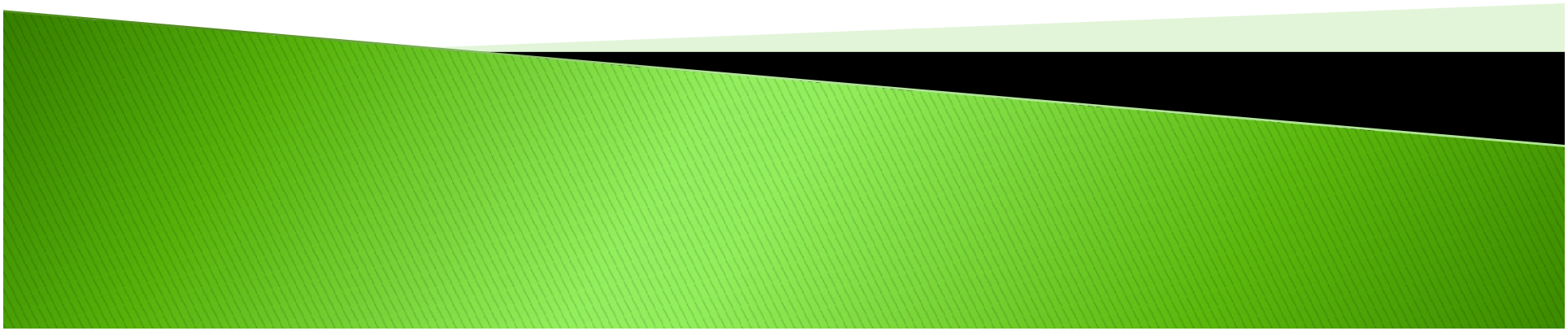
Solution.

The dual is

$$(A \cup \emptyset)^c \cup A = U$$



Finite Sets, Counting Principles

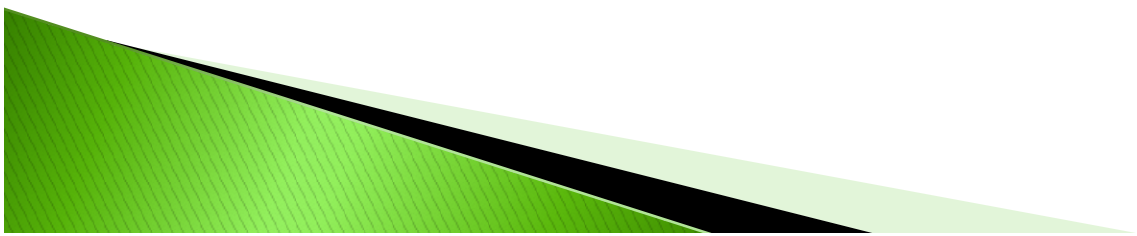


Finite Sets

A set is said to be finite if it contains exactly m distinct elements where m denotes some nonnegative integer. Otherwise set is said to be infinite. For example, the empty set ø and the set of letters of the English alphabet are finite sets, whereas the set of even positive integers $\{2, 4, 6, \ldots\}$ is infinite.

Notation

The notation $n(A)$ or $|A|$ will denote the number of elements in a finite set A.



Lemma 1

Suppose A and B are finite disjoint sets. Then $A \cup B$ is finite and

$$n(A \cup B) = n(A) + n(B)$$

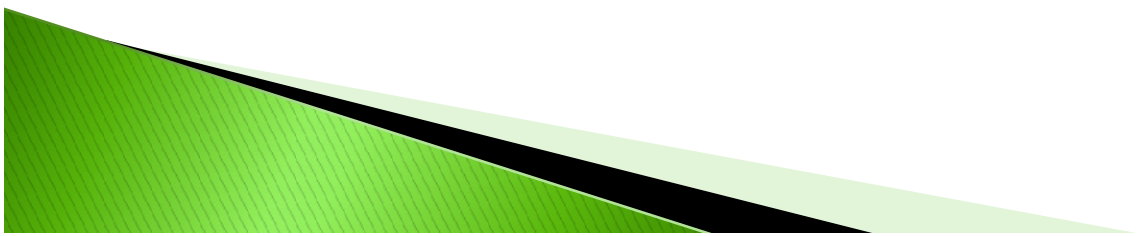
For example,

$A = \{1, 2, 3\}$ and $B = \{a, b, c\}$

Then,

$A \cup B = \{1, 2, 3, a, b, c\}$

$n(A \cup B) = 6 = n(A) + n(B) = 3 + 3 = 6$



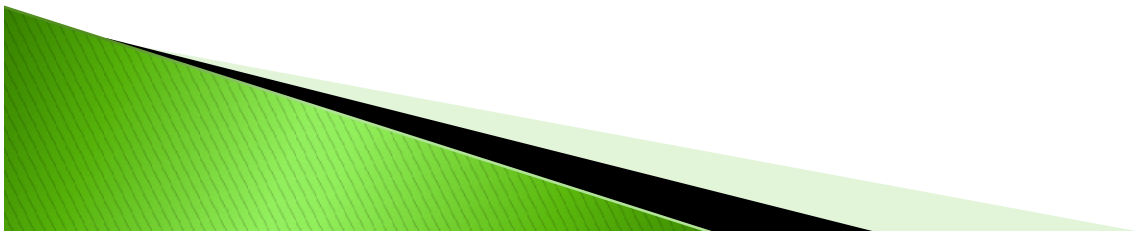
Theorem 1

Let A be any set in a finite universal set U. Then

$$n(A^c) = n(U) - n(A)$$

For example, if there are 20 male students in a class of 35 students, then there are 35-20=15 female students.

$n(U) - n(A)$



Theorem 2

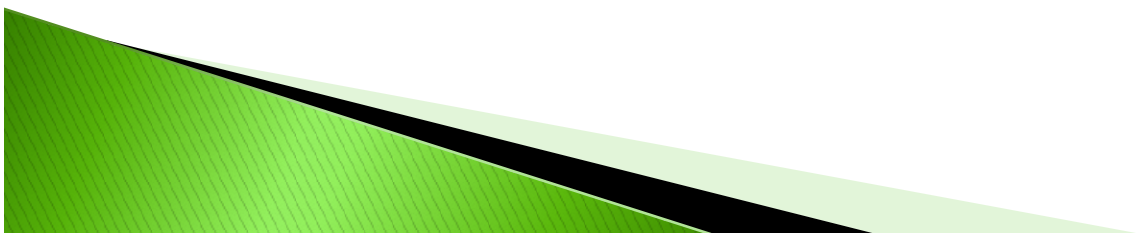
Suppose A and B are finite sets. Then

$$n(A \setminus B) = n(A) - n(A \cap B)$$

For example, suppose an archery class A contains 35 students, and 15 of them are also in a bowling class B. Then

$$n(A \setminus B) = n(A) - n(A \cap B) = 35 - 15 = 20$$

That is, there are 20 students in the class A who are not in class B.



Theorem 3

Suppose A and B are finite sets. Then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

For example,

$A = \{2, 4, 6, 8\}$, $B = \{6, 8, 10, 12\}$

Then, $A \cup B = \{2, 4, 6, 8, 10, 12\}$

$n(A \cup B) = 6$

$= n(A) + n(B) - n(A \cap B)$

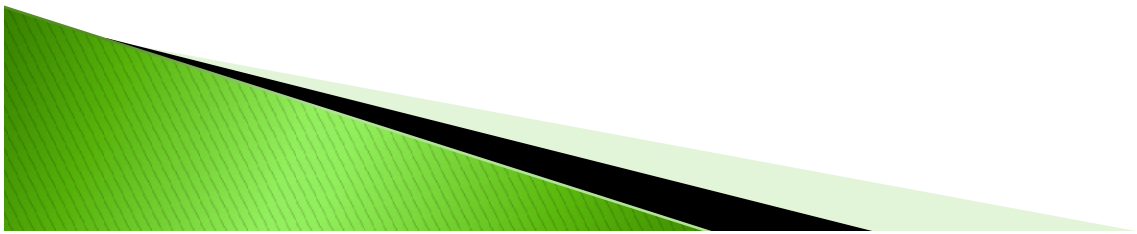
$= 4 + 4 - 2 = 6$

$A \cap B = \{6, 8\}$

Corollary 1

Suppose A, B, C are finite sets. Then $A \cup B \cup C$ is finite and

$$n(\mathrm{A} \cup \mathrm{B} \cup \mathrm{C}) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$



Example

Consider the following data among 110 students in a college dormitory:

30 students are on a list A (taking Accounting),

35 students are on a list B (taking Biology),

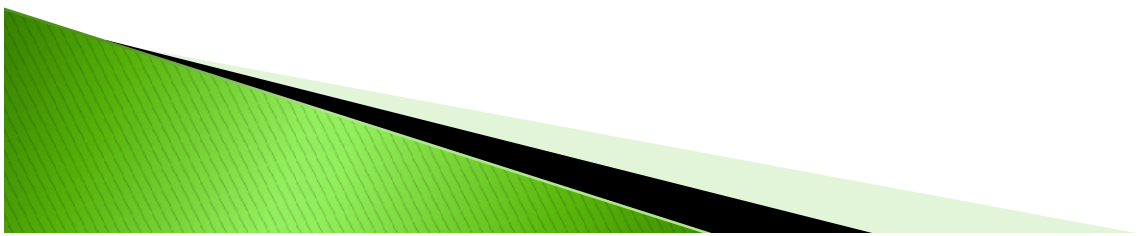
20 students are on both lists.

Find the number of students:

(a) On list A or B

(b) On exactly one of the two lists,

(c) On neither list.



Example

a) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 30 + 35 - 20 = 45$

b) $n(A/B) = n(A) - n(A \cap B)$

$= 30 - 20 = 10$

$n(B/A) = n(B) - n(A \cap B)$

$= 35 - 20 = 15$

c) $n(A^c \cap B^c) = n((A \cup B)^c)$

$= n(U) - n(A \cup B)$

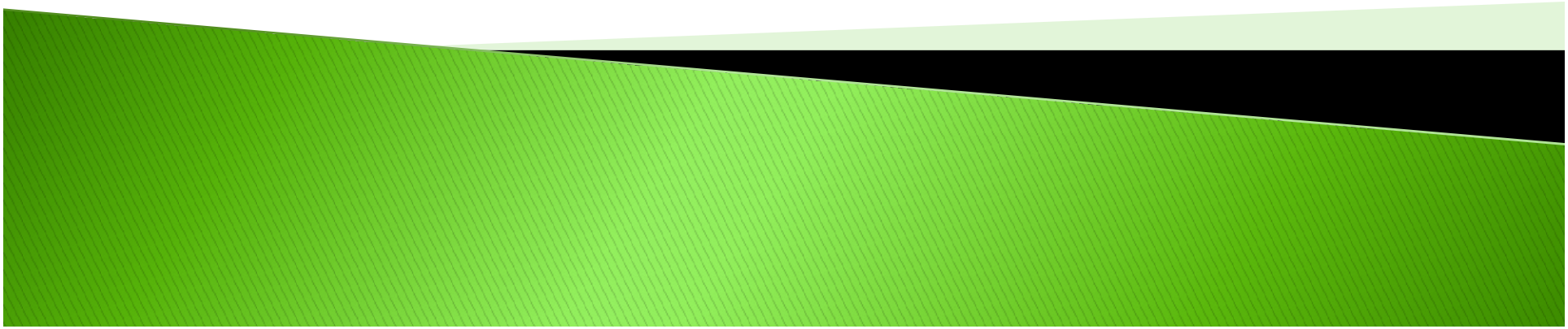
$= 110 - 45$

$= 65$

$(A \cup B)^c = A^c \cap B^c$

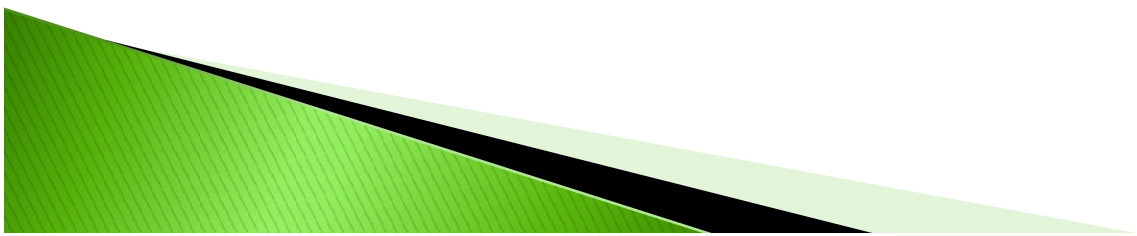
$\therefore n(A^c) = n(U) - n(A)$

Classes of Sets, Power Sets



Classes of Sets

Given a set *S*, we may wish to talk about some of its subsets. Thus we would be considering a "set of sets". Whenever such a situation arises, to avoid confusion, we will speak of *class* of sets or a *collection* of sets. If we wish to consider some of the sets in a given class of sets, then we will use the term *subclass* or *subcollection*.



Classes of Sets

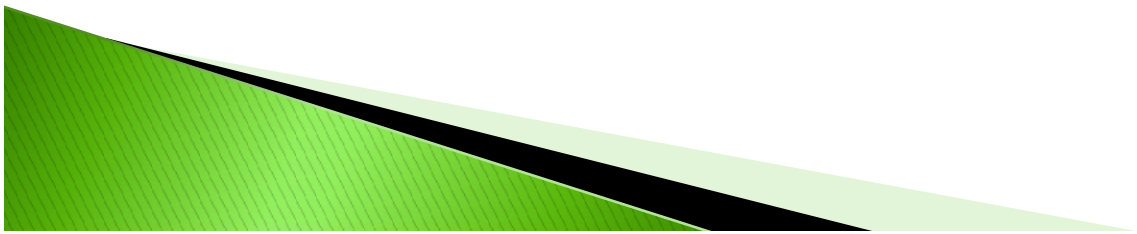
To understand this concept, we consider the set set $S = \{1, 2, 3, 4\}$. Let A be the class of subsets of S which contain exactly three elements of S. Then

$A = [\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}]$

Let B be the class of subsets of S which contain 2 and two other elements of S. Then

$B = [\{2,1,3\}, \{2,1,4\}, \{2,3,4\}]$

Here we can see that B is a subclass of A.

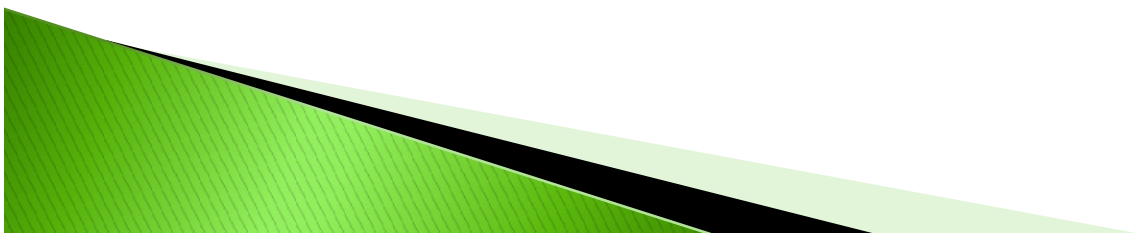


Power Sets

For a given set *S*, we may speak about the class of all subsets of S. This class is called the power set of *S*, and it will be denoted by *P(S)*. If *S* is finite, then so is *P(S)*.

Also

$$n(P(S)) = 2^{n(S)}$$



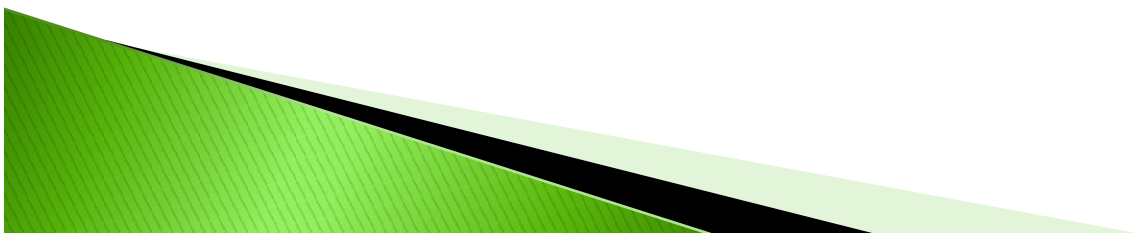
Power Sets

Suppose $S = \{1,2,3\}$. Then

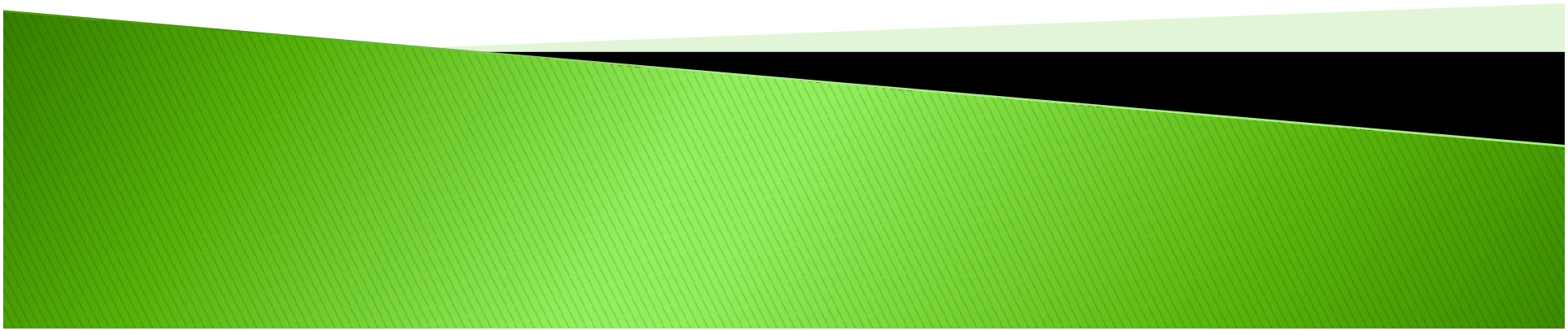
$$P(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$$

Note that

$$2^{n(S)} = 2^3 = 8 \text{ elements}$$



Mathematical Induction



Introduction

Suppose we have an infinite ladder, as shown in Figure 1.

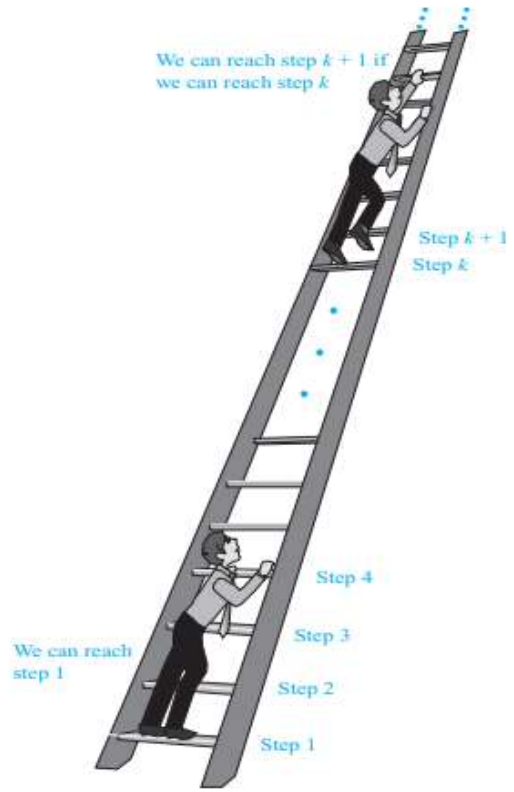


FIGURE 1 **Climbing an Infinite Ladder.**

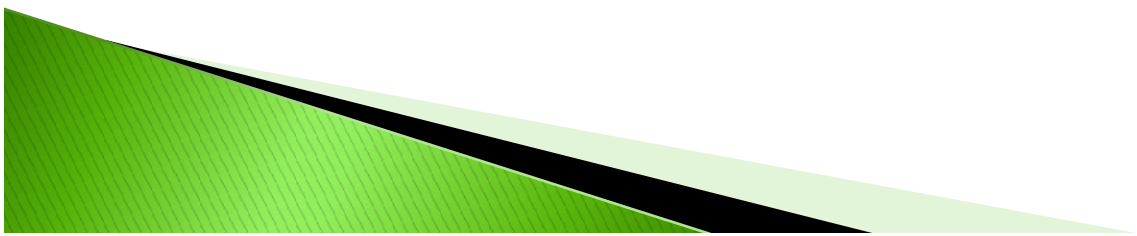
Introduction

We want to know whether we can reach every step on this ladder. We know two things:

1. We can reach the first step of the ladder.
2. If we can reach a particular step of the ladder, then we can reach the next step.

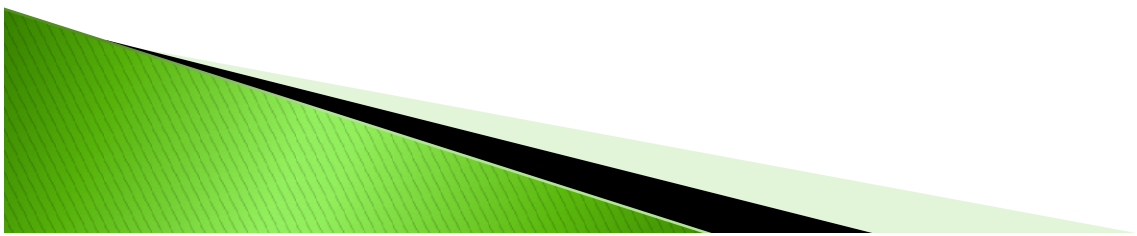
Can we conclude that we can reach every step?

Answer is yes. Something we can verify using an important proof technique called mathematical induction.



Mathematical Induction

In general, mathematical induction can be used to prove statements that assert that $P(n)$ is true for all positive integers n, where $P(n)$ is a proposition. A proof by mathematical induction has two parts, a basis step, where we show that $P(1)$ is true, and an inductive step, where we show that for all positive integers k, if $P(k)$ is true, then $P(k+1)$ is true.



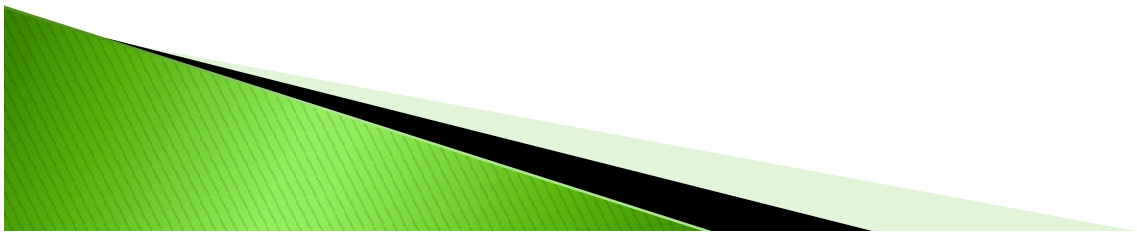
Principle of Mathematical Induction

To prove that $P(n)$ is true for all positive integers n, where $P(n)$ is a proposition, we complete two steps:

Basis Step: We verify that $P(1)$ is true.

Inductive Step: We show that the conditional statement $P(k) \longrightarrow P(k+1)$ is true for all positive integers k.

If $P(k)$ is true then we will prove that $P(k+1)$ is true.



Example 1

Show that if n is a positive integer, then

$$1+2+\ldots+n=\frac{n(n+1)}{2}$$

$1+2+3=\frac{3(3+1)}{2}$

$=\frac{12}{2}=6$

Let $P(n)$ be the proposition. First we show that $P(1)$ is true and then conditional statement $P(k)$ implies $P(k+1)$ is true for $k=1,2,3,\ldots$.

Basis step:

$P(1)$ is true: $1=\frac{1(1+1)}{2}$

$1=\frac{2}{2}=1$

Example 1

Inductive step: Assume that $P(k)$ is true for an arbitrary positive integer k

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$ ✓

Under this assumption it must be shown that $P(k+1)$ is true, namely

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)[(k+1)+1]}{2} = \frac{(k+1)(k+2)}{2}$$

Now, $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$

$$1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + k + 1$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

Example 1

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

This last equation shows that $P(k+1)$ is true under the assumption that $P(k)$ is true

Example 2

Let *P(n)* be the assertion that the sum of the first *n* odd integers is n^2 ; that is,

$$P(n): \quad 1+3+5+\ldots+(2n-1)=n^2.$$

$1+3+5 = 3^2$

$9 = 9$

Basis Step: P(1) is true, because

$1 = 1^2$

Inductive Step: Assume that P(k) is true. That is,

$1+3+5+\cdots+(2k-1)=k^2$

Next odd number after $2k-1$ is $2k+1$.

Example 2

$$1 + 3 + \cdots + (2k-1) = k^2 \checkmark$$

$$1 + 3 + \cdots + (2k-1) + (2k+1) = k^2 + (2k+1)$$

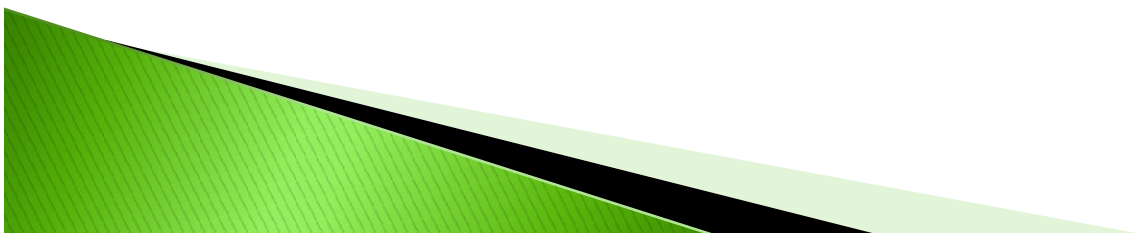
$$= \cancel{k^2 + 2k + 1}$$

$$= (k+1)^2 \checkmark$$

That is, $P(k+1)$ is true wherever $P(k)$ is true.

Proved

Example 2



Example 3

Use mathematical induction to show that

$$P(n): 2^0 + 2^1 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$$

for all nonnegative integers n.

$2^{(k+1)+1} - 1 = 2^{k+2} - 1$

Basis Step: Let $P(n)$ be the statement.

$P(0)$ is true, because

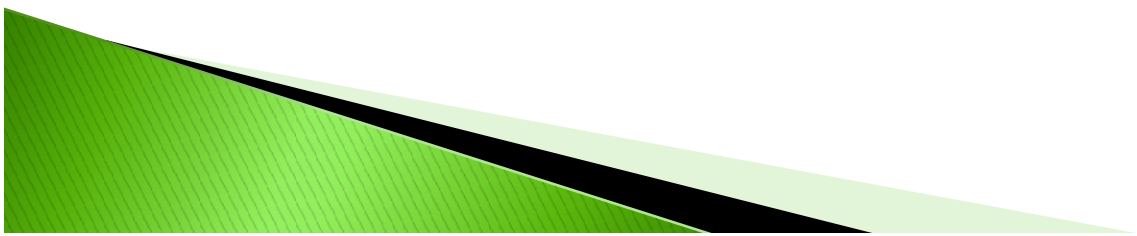
$2^0 = 2^{0+1} - 1$

$1 = 2^1 - 1$

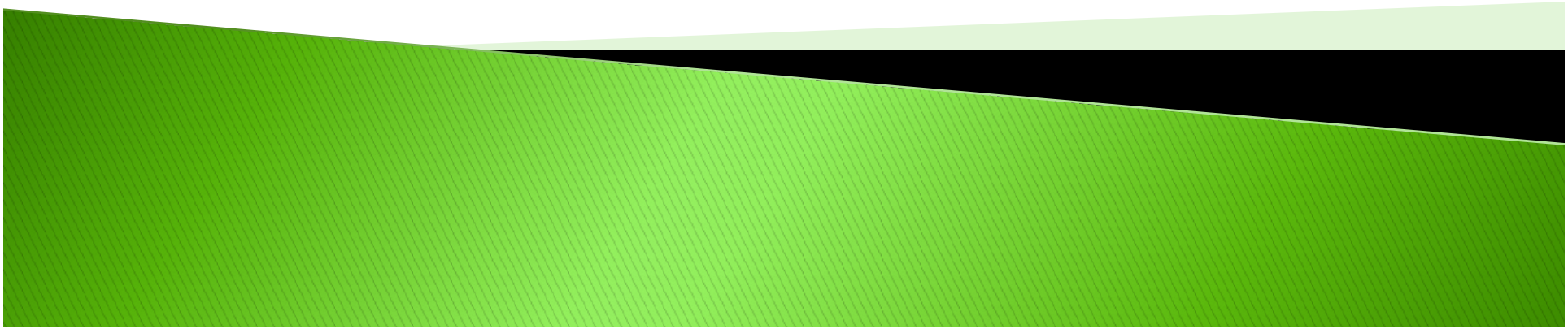
$1 = 1$

Inductive step: Assume that $P(k)$ is true. That $1 + 2 + 2^2 + \ldots + 2^k = 2^{k+1} - 1$ ✓

Example 3



Axiomatic Development of Set Theory

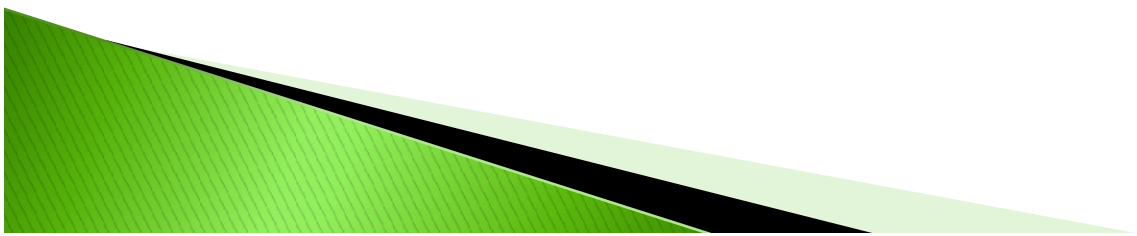


What is axiom and axiomatic development?

Any axiomatic development of a branch of mathematics begins with the following:

1) undefined terms,
2) Undefined relations,
3) Axioms relating the undefined terms and undefined relations.

Then one develops theorems based upon the axioms and definitions.



Axiomatic Development of Eulidean Geometry

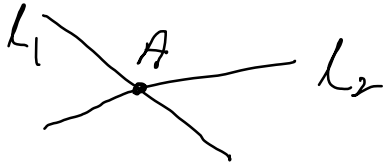
Consider, for example, the axiomatic development of plane Euclidean geometry. It begins with the following :

1) "points" and "lines" are undefined terms;
2) "point on a line" is an undefined relation.

Two of many axioms of Euclidean geometry follow:

Axiom 1: Two distinct points are on one and only one line.

Axiom 2: Two distinct lines cannot contain more than one point in common.



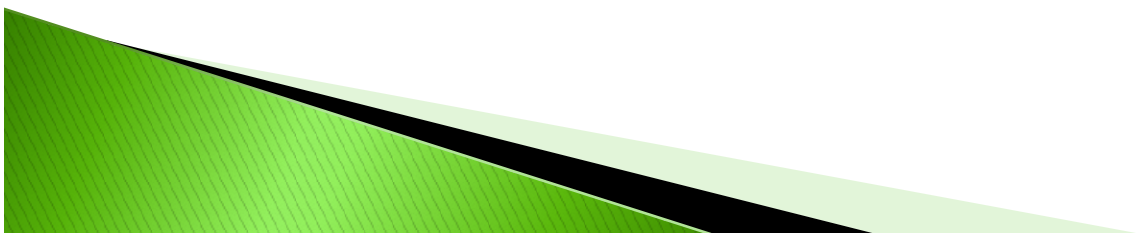
Axiomatic Development of Set Theory

The axiomatic development of set theory begins with the following:

1) "element" and "set" are undefined terms;
2) "elements belong to a set" is the undefined relation.

Two of the axioms of set theory follow:

Principle of Extension: Two sets *A* and *B* are equal if and only if they have the same elements.

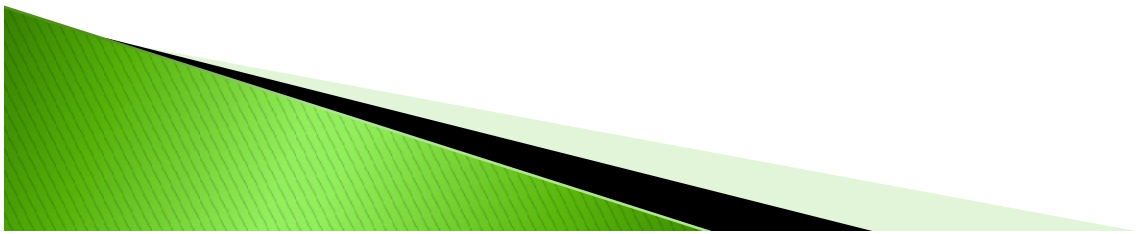


Axiomatic Development of Set Theory

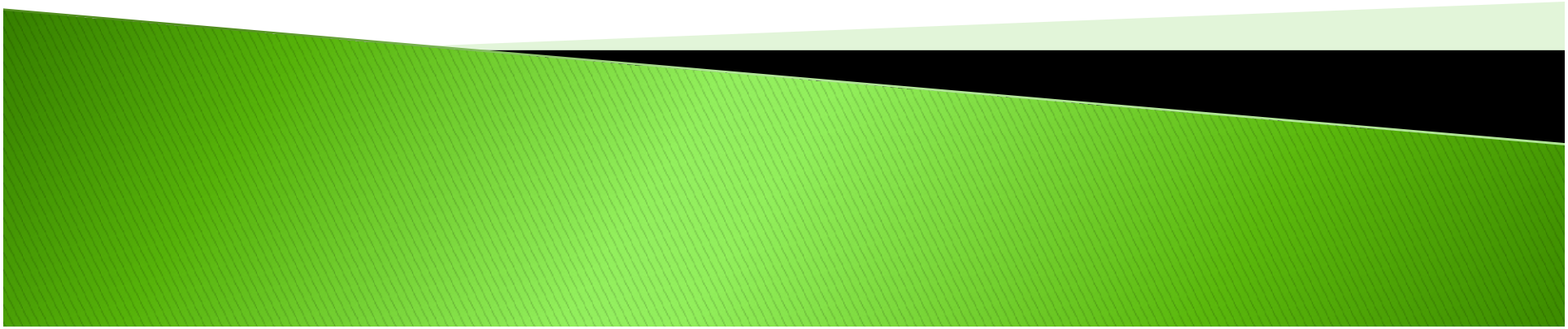
Principle of Abstraction: Given any set U and any property P, there is a set A such that the elements of A are exactly those elements in U which have the property P.

$U = \{$Nauman, Sara, Aleesha, Mansoor$\}$

$A = \{$Nauman, Mansoor$\}$



Real Number System



Real Number System

The notation $\underline{R}$ will be used to denote the real numbers. These are the numbers one uses in basic arithmetic and algebra. R together with its properties is called the real number system.

The set R of real numbers includes the following sets of numbers:

$Z = \{\ldots, -2, -1, 0, 1, 2, 3, \ldots\}$ = set of integers

$Z^+ = P = \{1, 2, 3, \ldots\}$ = set of positive integers

$N = \{0, 1, 2, 3, \ldots\}$ = set of nonnegative integers

Q = set of rational numbers i.e. numbers which are ratios of integers $\frac{2}{3}$, $-\frac{1}{2}$ $\frac{2}{1} = 2$

Real Number System

Irrational Numbers

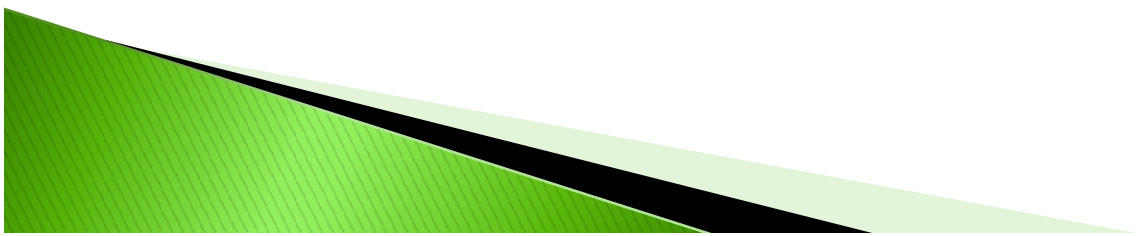
Those real numbers which cannot be represented as the ratio of integers, are called irrational numbers.

For example, $\pi = 3.14\ldots \neq \frac{22}{7}$

$\sqrt{2}$

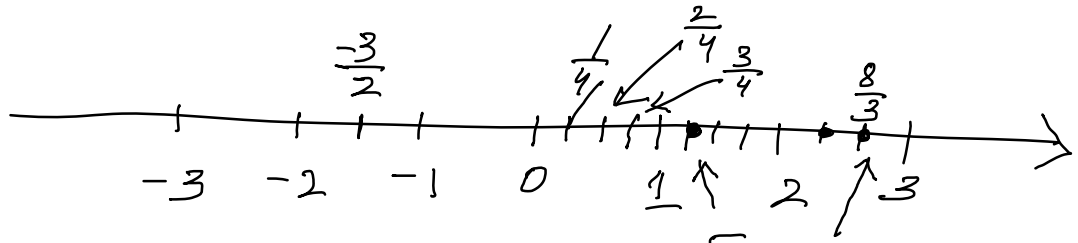
$\sqrt{3}$

$R = Q \cup Q'$



Real Line *R*, Decimal Expansion

One of the most important properties of the real numbers is that they can be represented graphically by points on a straight line.



$\frac{1}{4}$ ✓ $\quad \frac{5}{4} = 1\frac{1}{4}$ ✓

$\frac{8}{3} = 2\frac{2}{3}$ ✓

$$4\overline{)5}\ 1 \quad 4 \quad 1$$

$$3\overline{)8}\ 2 \quad 6 \quad 2$$

Decimal Expansion

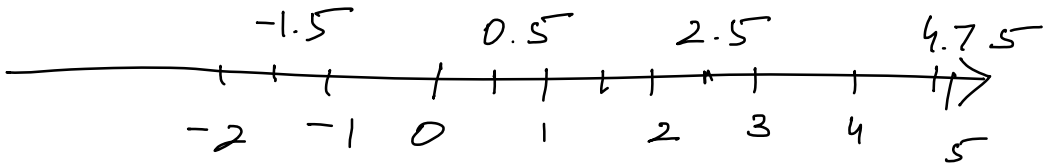
Real numbers can also be represented by decimals. The decimal expansion of a rational number will either stop as in ¾ = 0.75 or will have a pattern that repeats indefinitely, such as 17/11= 1.545454… Even when the decimal expansion stops, it can be rewritten using repeated 9`s, for example, ¾ = 0.74999….

$= 0.75$, $0.3 = 0.2999\ldots$

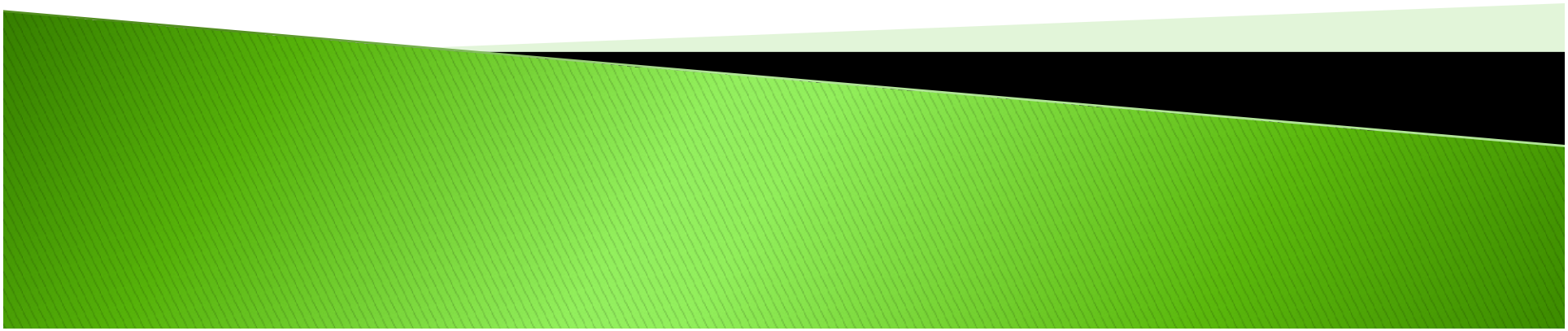
The decimal expansion of an irrational number never stops nor does it have a repeating pattern. The points representing the decimal 2.5 and 4.75 are indicated below on real line.

$\pi = 3.14\ldots$

Decimal Expansion



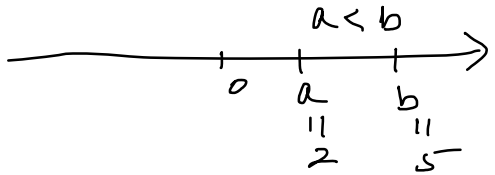
Order and Inequalities



Order Relation of Real Numbers

Let a and b be real numbers. We say a is less than b, written $a < b$ if the difference b-a is positive. Geometrically, $a < b$ if and only if the point a lies to the left of the point b on the real line R.

$2 < 5$
$5 - 2$ is positive



Observe that we define order in R in terms of the positive real numbers denoted by R^+.

Main Properties of Positive Real Number

There are two main properties of the positive real numbers R^+:

1) If a and b are positive, then $a+b$ and ab are positive.
2) For any real number a, either a is positive, $a = 0$, or $-a$ is positive.

$a = 2$ $\quad -a = -(-2) = 2$

$a = -2$ $\quad a = 0$

The following additional notation and terminology are used:

$a > b$, means $b < a$

$a \leq b$, means $a < b$ or $a = b$ $\quad 2 \leq 2$ or $2 = 2$

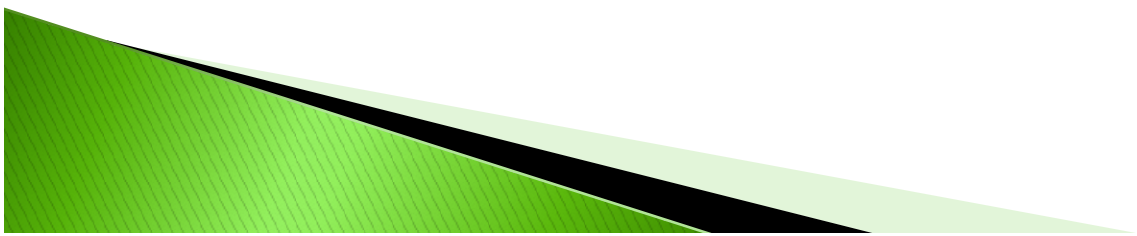
$a \geq b$, means $b \leq a$ $\quad 2 < 5$ or $5 > 2$.

Inequality

Any statement of the form $a < b$, $a \leq b$, $a > b$, *or* $a \geq b$ is called inequality; and any statement of the form $a < b$ or $a > b$ is sometimes called a strict inequality.

$2 < 7$ ✓ strict inequality

$a \geq b$ × not strict inequality

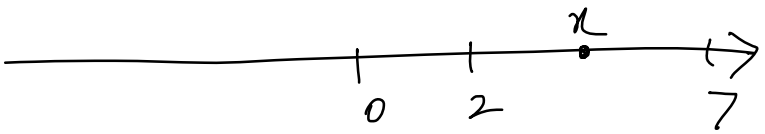


Example 1

a) $2 < 5$; $-6 < -3$, $4 \leq 4$, $5 > -8$, $6 \geq 0$, $-7 \leq 0$

b) A real number a is positive iff $a > 0$, and a is negative iff $a < 0$.

c) The statement $2 < x < 7$ means $2 < x$ and $x < 7$.



Properties of Inequality

Basic properties of the inequality relation follow.

1) Let a, b, c be real numbers. Then:

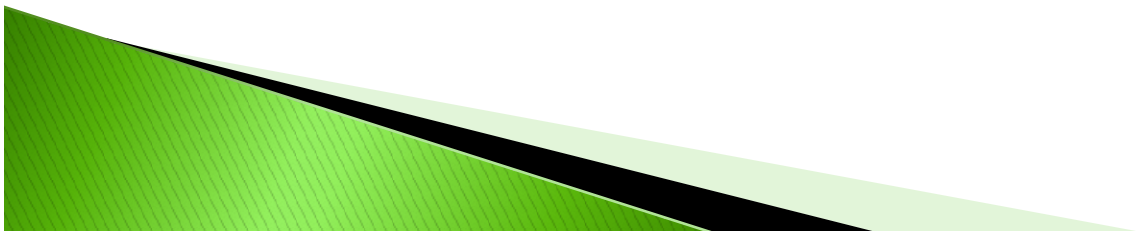
   (i) $a \leq a$ $2 \leq 2$

   (ii) If $a \leq b$ and $b \leq a$, then $a = b$! $3 \leq 3, \Rightarrow 3=3$

   (iii) If $a \leq b$ and $b \leq c$, then $a \leq c$. $2<5, 5<7 \Rightarrow 2<7$

2) For any real numbers a and b, exactly one of the following holds:

   $a < b$, $a = b$ or $a > b$



Properties of Inequality

3) Let a, b, c be real numbers such that $a \leq b$. Then

(i) $a+c \leq b+c$.

(ii) $ac \leq bc$ when $c > 0$; but $ac \geq bc$ when $c < 0$.

$a = 2, \quad b = 5, \quad c = 2$

$a < 5 = b$

$a + c < b + c$

$4 < 7$

$a < b$

$ac < bc$

$4 < 10$

let $c = -2$

$ac > bc$

$-4 > -10$

Properties of Inequality

Q. Prove that if $a \leq b$ and $b \leq c$, then $a \leq c$.

Proof: If $a = b$ and $b = c$, then $a = c$.

If $a < b$ and $b < c$, then $b - a$ is positive and $c - b$ is positive.

Now, $(b - a) + (c - b)$ is positive.

$= -a + c$

$= c - a$ is positive

$\Rightarrow a < c$

Hence, $a < c$. Proved

Properties of Inequality

Q. Let a, b, c be real numbers. Then prove that and suppose $a \leq b$

(i) $a+c \leq b+c$.

(ii) $ac \leq bc$ when $c > 0$; but $ac \geq bc$ when $c < 0$.

Do it yourself as assignment.

If $a = b$, then $ac = bc$

If $a < b$, then $b - a$ is positive

$c(b-a)$ is also positive.

$= bc - ac$ is " "

$\Rightarrow ac < bc$

Hence $ac \leq bc$